

## TRADITIONAL RAN

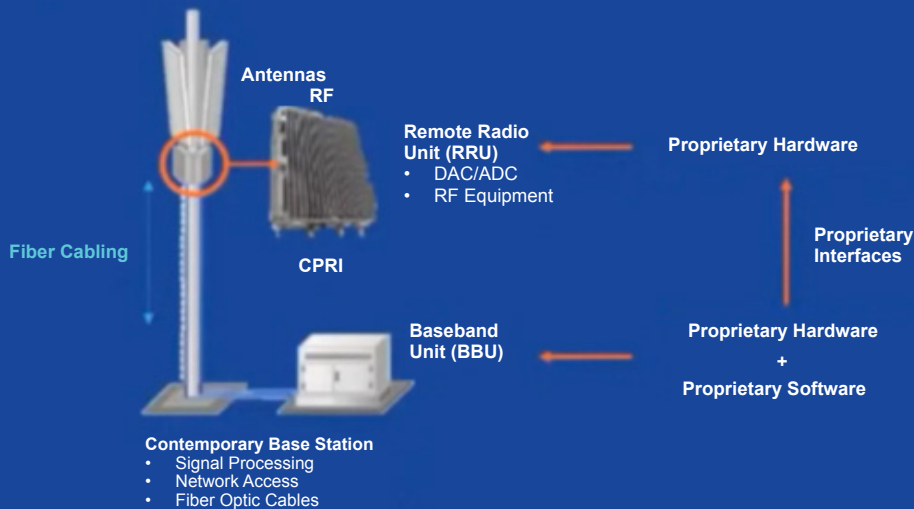


Figure 3: Traditional RAN

## OPEN RAN: DISAGGREGATING HARDWARE AND SOFTWARE

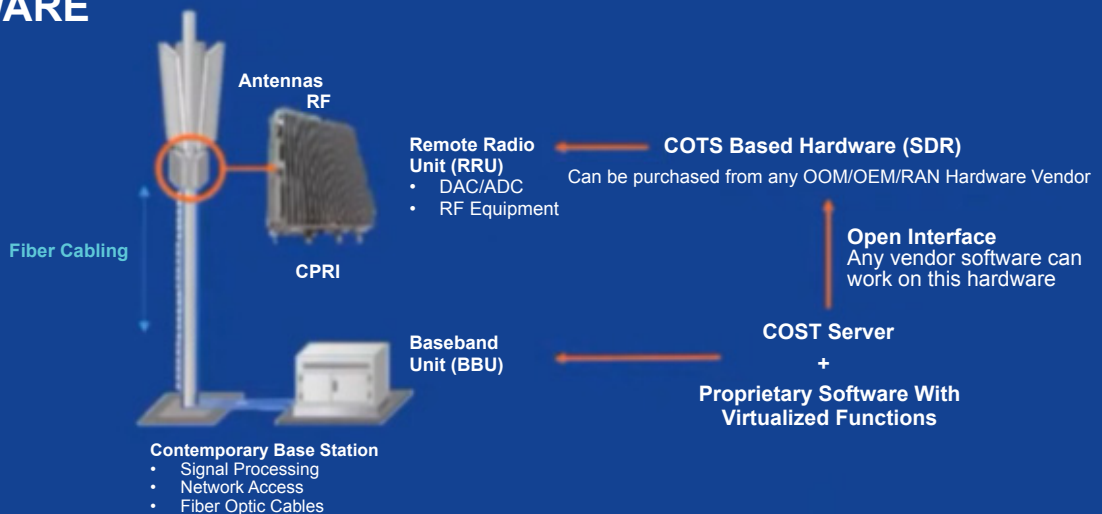


Figure 4: Open RAN architecture

To have a more detailed understanding of this architecture, let us understand a few things about previous generation network architectures first. All the electronics was in one or more cabinets at the bottom of a mast (tower) when 2G and 3G were in use. The RF cables would run to the antennas on top of the mast. The RF signal continuation in the RF cable was a big problem because

of the congestion that used to occur. 3G and 4G allowed the industry to take a different approach that worked better. Instead of sending RF signals from the cabinet to the antennas, they were moved closer to the antennas (as can be seen in Figure 3). This ensured that there was a very minimal loss of signals.

This new piece of electronics is known as remote radio unit (RRU) or remote radio head (RRH) — at the

bottom is the baseband unit. A cabinet consists of both the hardware and the software. RRU is always proprietary hardware, whereas BBU is proprietary software along with the interfaces between BBU and RRU.

### Understanding the Open RAN architecture

In Open RAN, the RRU is COTS based hardware that can be